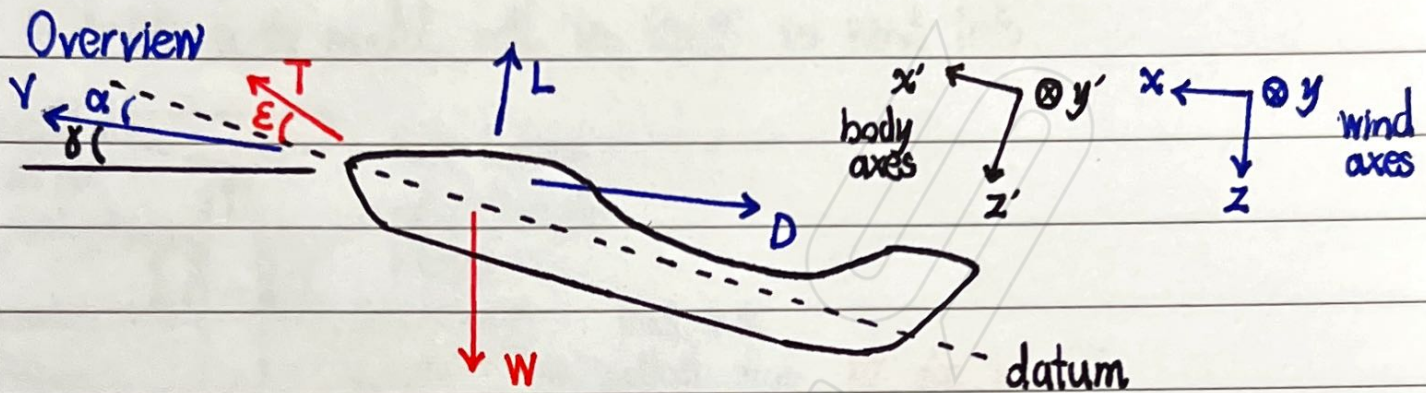


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Mathematical model

↳ Predicted performance and behaviour ← uncertainty

↳ Actual performance and behaviour ← uncertainty

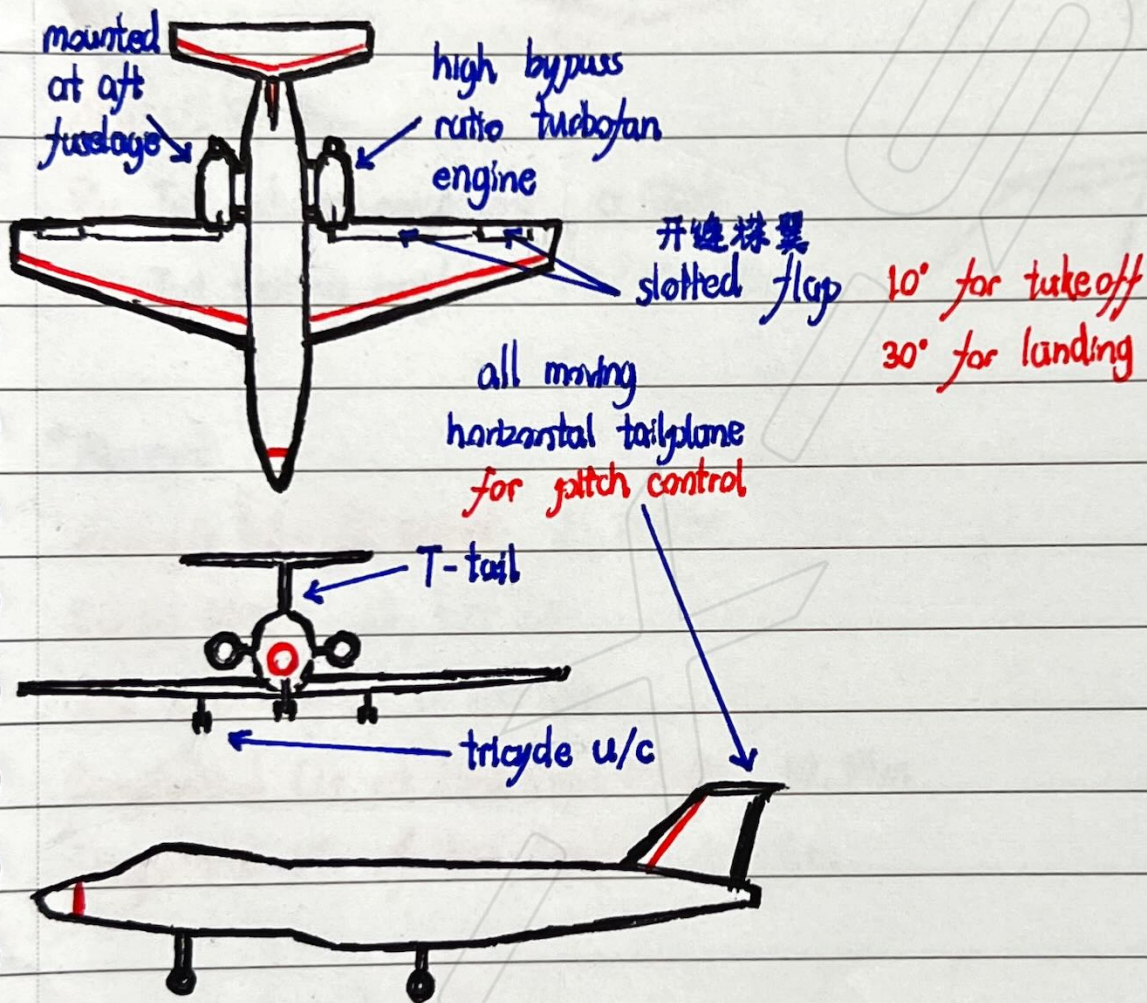
- actual stall speed
- rate of climb
- turn performance
- max speed profile
- mission performance
- field performance
- stability characteristics
- ↳ response to pilot inputs

↳ Lengthy flight-testing programs

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The Aircraft

* 1:24 scale model will be used in next lab

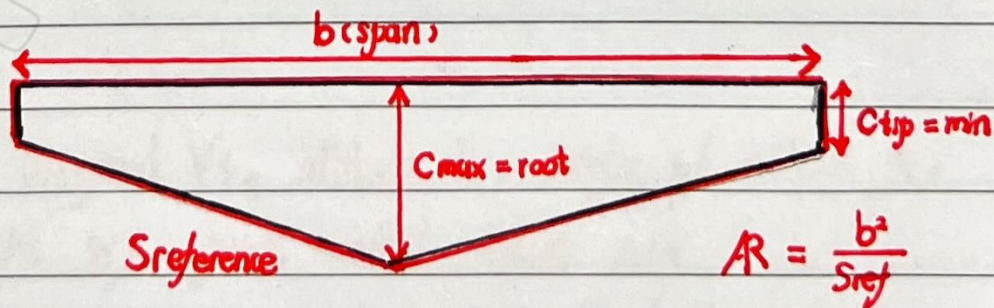


* Wing

S_{ref} 13.73 m²

$\bar{c}$ 3.79 m

AR 5.93



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* Horizontal tail

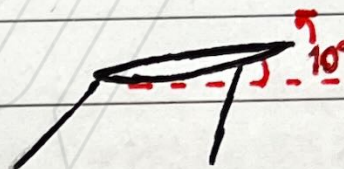
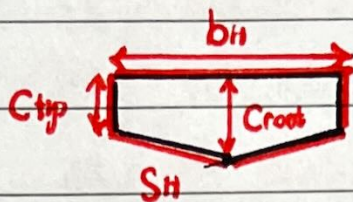
S_H 18.43 m²

$\bar{C}_H$ 1.97 m

R_H 4.74

$\bar{V}_H$ Tail volume coefficient 0.6305

Δi_H Tail setting angles +5/-10 deg



* Aircraft

Maximum Take off Weight 24,348 lbs

Empty Weight 13,434 lbs

Max Fuel Weight 8,800 lbs

Longitudinal CG at Zero-Fuel Weight 10.14 m

Longitudinal CG of fuel tanks 10.06 m

Setup & Procedure

- act as pilot
- act as engineer

→ record indicated airspeed V_i , altitude h , angle of attack α , fuel weight W_f , pitch control input Δi_H

* MAKE AN EXCEL SHEET! ** Use Test Cards!

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- ↳ data → determine aircraft's lift curve ($\alpha, \alpha_0, C_{Lmax}$)
drag polar (C_{D0}, k)
zero-lift pitching moment (C_{M0})
neutral point (X_{np})

① Stall test

- $h = 4000 \sim 4500\text{ft}$
↳ flying aircraft level $V \geq 150\% V_s \rightarrow V \approx 140\text{kn}$ IAS 指示空速
↳ engine throttle → ideal → slowly pull back on the controls $\alpha \uparrow$
↳ maintain level flight → $V \downarrow$
↳ $\hat{=}$ stall warning → keep 2 — until — "7" buffeting of the controls
further $\uparrow$ of α OR → no further control authority ✗
* record V_s ** record any wing drop tendency $\rightarrow$ ~~30~~ 5
↳ stall recovery procedure — same time → full throttle
↳ -5° nose down attitude

* x flaps (flaps up)

x undercarriage (u/c up)

— first attempt

✓ landing configurations 30° flaps + u/c down second attempt

At least two measurement for each configuration !!!

② Glide test

Test rate of descent at different V

determine $(\frac{P}{W})$ vs. h and V

$$\rightarrow \frac{dh}{dt} = V \left(\frac{T}{W} - \frac{1}{L/D} \right) = \eta_p \frac{P}{W} - \frac{V}{L/D}$$

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↳ When V constant in glide $\rightarrow \frac{P}{W} = \frac{T}{W} = 0 \rightarrow L/D$ by $\frac{dh}{dt}$

↳ climbing $h = 2000\text{ft} > \text{test altitude if possible } 4500\text{ft}$

↳ throttle $\rightarrow$ ideal

↳ attain test air speed

↳ keep speed constant as possible

* $V_{ias} \downarrow \leftarrow$ nose up + release some back pressure (for 4)

$V_{ias} \uparrow \leftarrow$ nose down + apply some back pressure (for 4)

↳ measure time

$$t = h_{\text{test}} + 500\text{ft} \rightarrow h_{\text{test}} - 500\text{ft}$$

Repeat by different V_{ias} !!! Repeat with same V_{ias} if possible !

③ Steady Level Flight

Test lift curve slope, tail lifting characteristics, zero-lift pitching moment and static margin by estimate through steady flight with different V and CG

↳ required amount of fuel $\rightarrow$ set weight

to achieve correct CG offset for zero-fuel CG position

+6, -6 and 0 from the instructor's console

↳ climb/descend $h = h_{\text{desired}} = 5500\text{ft}$

↳ throttle $\rightarrow$ full

adjust stick input to maintain altitude

$a \uparrow \rightarrow$ nose down $\uparrow$

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↳ when V constant, $V = 100, 150, 200, 250 \dots$

record air speed, tailplane angle, α and m_{fuel}

③.1 test lower air speed 1

↳ throttle → reduce by 20% → decelerate → $V = V_{desired}$

↳ maintain altitude adjust stick input

- $\alpha \uparrow$ → nose up $\uparrow$

↳ throttle → increase → until V constant when reach IAS

↳ when V constant

record air speed, tailplane angle, α and m_{fuel}

↳ Repeat 1-4 for $V \downarrow$

③.2 test lower air speed 2 (easier)

↳ climb until reach IAS

↳ maintain IAS required adjust stick

↳ throttle → reduce slowly → maintain TAS till flying level

↳ record air speed, tailplane angle, α and m_{fuel}

↳ Repeat 1-4 for $V \downarrow$

Completing the Lab

① safety

② Conduct of the lab

4 stall test + 5~6 glide test + 12 level flight test (4V + 3CG)

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③ Take-off

1 → throttle → full, flaps 10°

2 → maintain centre line → rudder pedal

3 → $V = 90 \text{ knots}$ → pull back → climb angle 10°

4 → $V = 120 \text{ knots}$ → raise V/C → raise flaps

5 → $h = h_{\text{desired}}$

throttle → $\approx 60\%$

adjust pitch to maintain altitude

3/1/2023 Flight Simulator During Lab

Pilot:		Date:		Time:	
CG offset (in):		Empty Weight:			
Test Altitude (ft):		Payload Weight:			
Check	Action				
	GLIDE TEST - POWER-OFF, WINGS LEVEL				
	Set aircraft to clean configuration				
	Ascend to 2000 ft above test altitude				
	Adjust throttle & pitch to maintain altitude at test speed				
	Set throttle to idle				
	Pitch down to maintain test speed				
	Start timer 500 ft over test altitude				
	Maintain test speed				
	Record time 500 ft below test altitude				
	IAS (kn)		Fuel Weight:		
	Time to Descend:				
	IAS (kn)		Fuel Weight:		
	Time to Descend:				
	IAS (kn)		Fuel Weight:		
	Time to Descend:				
	IAS (kn)		Fuel Weight:		
	Time to Descend:				

Notes:

3/2/2023 Flight Simulator During Lab

3/2/2023 Flight Simulator During Lab

Pilot:		Date:		Time:	
CG offset (in):		Empty Weight:			
Test Altitude (ft):		Payload Weight:			
Check	Action				
	GLIDE TEST - POWER-OFF, WINGS LEVEL				
	Set aircraft to clean configuration				
	Ascend to 2000 ft above test altitude				
	Adjust throttle & pitch to maintain altitude at test speed				
	Set throttle to idle				
	Pitch down to maintain test speed				
	Start timer 500 ft over test altitude				
	Maintain test speed				
	Record time 500 ft below test altitude				
	IAS (kn)		Fuel Weight:		
	Time to Descend:				
	IAS (kn)		Fuel Weight:		
	Time to Descend:				
	IAS (kn)		Fuel Weight:		
	Time to Descend:				
	IAS (kn)		Fuel Weight:		
	Time to Descend:				

Notes:

Notes:

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Notes:

Notes: 4

3/2/2023 Flight Simulator During Lab

Notes:

Notes:

3/2/2023 Flight Simulator During Lab

Pilot:		Date:		Time:	
CG offset (in):		Empty Weight:			
Test Altitude (ft):		Payload Weight:			
Check	Action				
TRIM TEST					
	Climb/Descend to the test altitude				
	Adjust pitch & throttle to fly level flight at desired IAS				
	Record data				
	Test 1				
	IAS (kn)		Fuel Weight:		
	Stabilator (deg)		AoA (deg):		
	Test 2				
	IAS (kn)		Fuel Weight:		
	Stabilator (deg)		AoA (deg):		
	Test 3				
	IAS (kn)		Fuel Weight:		
	Stabilator (deg)		AoA (deg):		
	Test 4				
	IAS (kn)		Fuel Weight:		
	Stabilator (deg)		AoA (deg):		
	IAS (kn)		Fuel Weight:		
	Stabilator (deg)		AoA (deg):		

Notes:

3/2/2023 Flight Simulator During Lab

Notes:

3/2/2023 Flight Simulator During Lab

Pilot:		Date:		Time:	
CG offset (in):		Empty Weight:			
Test Altitude (ft):		Payload Weight:			
Check	Action				
	GLIDE TEST - POWER-OFF, WINGS LEVEL				
	Set aircraft to clean configuration				
	Ascend to 2000 ft above test altitude				
	Adjust throttle & pitch to maintain altitude at test speed				
	Set throttle to idle				
	Pitch down to maintain test speed				
	Start timer 500 ft over test altitude				
	Maintain test speed				
	Record time 500 ft below test altitude				
	IAS (kn)		Fuel Weight:		
	Time to Descend:				
	IAS (kn)		Fuel Weight:		
	Time to Descend:				
	IAS (kn)		Fuel Weight:		
	Time to Descend:				
	IAS (kn)		Fuel Weight:		
	Time to Descend:				

Notes:

Notes: Stall test failed as the V too high

3/2/2023 Flight Simulator During Lab

Pilot:		Date:		Time:	
CG offset (in):		Empty Weight:			
Test Altitude (ft):		Payload Weight:			
Check	Action				
	GLIDE TEST - POWER-OFF, WINGS LEVEL				
	Set aircraft to clean configuration				
	Ascend to 2000 ft above test altitude				
	Adjust throttle & pitch to maintain altitude at test speed				
	Set throttle to idle				
	Pitch down to maintain test speed				
	Start timer 500 ft over test altitude				
	Maintain test speed				
	Record time 500 ft below test altitude				
	IAS (kn)		Fuel Weight:		
	Time to Descend:				
	IAS (kn)		Fuel Weight:		
	Time to Descend:				
	IAS (kn)		Fuel Weight:		
	Time to Descend:				
	IAS (kn)		Fuel Weight:		
	Time to Descend:				

Notes:

3/2/2023 Flight Simulator During Lab

Pilot:		Date:		Time:	
CG offset (in):		Empty Weight:			
Test Altitude (ft):		Payload Weight:			
Check	Action				
TRIM TEST					
	Climb/Descend to the test altitude				
	Adjust pitch & throttle to fly level flight at desired IAS				
	Record data				
	Test 1				
	IAS (kn)		Fuel Weight:		
	Stabilator (deg)		AoA (deg):		
	Test 2				
	IAS (kn)		Fuel Weight:		
	Stabilator (deg)		AoA (deg):		
	Test 3				
	IAS (kn)		Fuel Weight:		
	Stabilator (deg)		AoA (deg):		
	Test 4				
	IAS (kn)		Fuel Weight:		
	Stabilator (deg)		AoA (deg):		
	Test 5				
	IAS (kn)		Fuel Weight:		
	Stabilator (deg)		AoA (deg):		

Notes:

3/2/2023 Flight Simulator During Lab